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Run and Test Guide

How to run the project and check every case: one command, automated suites, or by hand

This guide shows how to start the app and how to check every approval case, in three ways that you can use separately or together:

Way	Effort	What it proves
A. One command (`scripts/try_all_cases.py`)	30 seconds	All 25 cases pass against a real running server
B. Automated test suites (pytest, mutation, Playwright)	2 minutes	The unit, API and UI tests pass; the tests can detect bugs
C. By hand in the web UI	15 minutes	You can see each rule working, with your own eyes

1. Before you start

You need Python 3.10 or newer (3.13 was used) and a terminal in the repository folder (the one containing `backend`, `e2e`, `scripts`, `docs`). Node.js is needed only for the Playwright test, and WSL with Ubuntu 22.04 only for the mutation test.

Install once:

```
pip install -e "backend[dev]"
```

2. Run the project

```
python -m uvicorn invoice_app.api.main:app --app-dir backend/src --port 5000
```

Wait for the line `Application startup complete`, then open http://localhost:5000 in a browser. You should see a **Sign in** form. Stop the server with `Ctrl+C`.

Important facts:

- **Data lives in memory.** Every restart resets everything to the demo data. Restart whenever you want a clean slate.
- **Port busy?** Use another port: `--port 5001`, then open `http://localhost:5001`.
- The OpenAPI page at `/docs` lists the endpoints, but the calls need a sign-in token, so use the UI or the script for real testing.

Demo accounts

All passwords are `ChangeMe123!`. These are local demo accounts, not real credentials.

Username	Role	Can approve?
`demo.manager`	Manager	Up to 50,000.00; first approval above 500,000.00

Username	Role	Can approve?
`demo.director`	Director	Up to 500,000.00 alone; either approval above that
`demo.director2`	Director	Same as above (a second, different Director)
`demo.admin`	Admin	Never (can reject and delete only)

Demo invoices already in the app

Invoice	Amount	Status	Good for
INV-1001, 1002, 1004	4,250.00; 980.50; 615.75	Pending	Simple approve and reject
INV-1003	12,300.00	Approved	Already finished
INV-1005	310.00	Draft	Needs "Submit for approval" first
INV-2001	75,000.00	Pending	Manager refused, Director allowed
INV-2002	600,000.00	Pending	Needs two different approvers

3. Way A: try every case with one command

With no server running, from the repository folder:

```
python scripts/try_all_cases.py
```

The script starts its own fresh server (on port 5058), runs all 25 cases, prints one line each, stops the server, and exits with code 0 only if every case passed. Each case creates its own invoice, so the cases do not depend on each other. To use a server that is already running instead:

```
python scripts/try_all_cases.py --url http://localhost:5000
```

Expected last line: `25 of 25 cases passed`. The script also checks the **exact reason text** of every refusal and that **the invoice is identical before and after** a refused attempt.

The 25 cases and their result (last run: all passed)

ID	Case	Expected outcome	Result
C01	Manager approves exactly 50,000.00	200, Approved, decided by the Manager	PASS
C02	Manager approves 50,000.01 (one cent over)	403 Manager-limit reason; invoice unchanged	PASS
C03	Director approves 50,000.01	200, Approved with one approval	PASS

ID	Case	Expected outcome	Result
C04	Director approves exactly 500,000.00	200, Approved; no second approver needed	PASS
C05	Manager approves exactly 500,000.00	403 Manager-limit reason; invoice unchanged	PASS
C06	Admin approves a small invoice	403 "Role Admin is not allowed..."; unchanged	PASS
C07	Admin approves an invoice above 500,000	403 role reason; unchanged	PASS
C08	Director gives the first approval on 500,000.01	200; still Pending Approval; "1 of 2"; no decider	PASS
C09	The same Director approves again	403 duplicate-user reason; still 1 approval	PASS
C10	The Manager gives the second approval after a Director	403 "second approver must be a Director"; still 1 approval	PASS
C11	A different Director gives the second approval	200, Approved, 2 approvals, decided by the second Director	PASS
C12	Manager first, then a Director, on 500,000.01	200, Approved, 2 approvals	PASS
C13	The same Manager approves twice on 500,000.01	403 duplicate-user reason; still 1 approval	PASS
C14	A refused attempt, then a valid second approver	The valid approver still completes the invoice	PASS
C15	Approve an already Approved invoice	409 state error; unchanged	PASS
C16	Approve a Draft invoice	409 state error	PASS
C17	Approve an invoice that does not exist	404 not found	PASS
C18	Reject a pending invoice with a reason	200, Rejected, reason stored	PASS
C19	Reject without a reason	400; invoice stays pending	PASS
C20	Reject an invoice that already has its first approval	200, Rejected; first approval kept on record	PASS
C21	Approve without signing in	401	PASS
C22	Sign in with a wrong password	401 "Invalid username or password."	PASS
C23	Create invoices with 0, -5, 50000.001, 1e999999, NaN	400 each; nothing stored; list still works	PASS
C24	Delete as Manager, then as Admin	403 for the Manager, 204 for the Admin	PASS
C25	List filtered by status; unknown status	Only Approved invoices; 400 for an unknown status	PASS

How we know the script can fail

The script was also run against a deliberately broken rule (`<` changed to `<=` in the Manager limit rule). It then reported `24 of 25 cases passed` and failed C01 with the message "A Manager may approve invoices up to 50,000.00; this invoice is 50,000.00 and needs a Director." The file was restored and the run went back to 25 of 25.

4. Way B: the automated suites

Run each from the repository folder.

B1. Unit and API tests (about 1 second)

```
python -m pytest backend/tests -q
```

Expected: `177 passed`.

Module	Tests	What it covers
`test_approval_policy.py`	63	The pure rules: limits, boundaries, roles, the two negative rules, each rule class on its own
`test_approval_service.py`	62	The lifecycle, persistence, refusal leaves nothing changed, state errors, concurrency, injected policy and clock
`test_api.py`	36	Status codes, messages, sign-in, validation, JSON shape, injected collaborators
`test_repository_contract.py`	16	One contract that the fake and the in-memory repository must both pass

To run only the two negative rules, verbosely:

```
python -m pytest backend/tests/test_approval_policy.py -v -k "same_user or second_approver"
```

B2. Which tests prove which rule

Rule	Where it is tested
Manager up to and including 50,000	`test_manager_may_approve_up_to_and_including_the_manager_limit`; service test at exactly the limit; API test for a small invoice
Manager refused above 50,000	`test_manager_may_not_approve_above_the_manager_limit`; `test_manager_is_refused_one_cent_above_the_manager_limit_and_nothing_changes`; API 403 test

Rule	Where it is tested
Director up to and including 500,000	`test_director_may_approve_up_to_and_including_the_director_limit`; `test_director_approves_alone_at_exactly_the_director_limit`
Other roles never approve	`test_roles_other_than_manager_and_director_may_not_approve_anything`; Admin refusal tests at service and API level
Above 500,000 needs two approvals	`test_required_approvals_steps_up_only_above_the_director_limit`; `test_first_approval_above_the_director_limit_keeps_the_invoice_pending`; `test_second_distinct_director_completes_the_approval`
Negative rule 1: same user never both approvers	`test_same_user_cannot_be_both_approvers`; `test_same_user_is_recognised_despite_case_and_whitespace`; stored-side variants; `test_same_director_cannot_approve_twice_and_the_invoice_keeps_exactly_one_approval`; API `test_same_director_twice_gets_403_and_the_first_approval_stays_alone`
Negative rule 2: second approver must be a Director	`test_second_approver_who_is_a_manager_is_refused_with_a_reason`; `test_a_manager_as_second_approver_is_refused_and_nothing_changes`; API `test_manager_as_second_approver_gets_403`
Refusal changes nothing	Helper `assert_refused_and_unchanged` (reason text, invoice equals its copy, repository saw no save); `test_a_refused_approval_is_never_persisted`; API tests comparing the full invoice before and after
Two simultaneous approvals by one user	`test_two_simultaneous_approvals_by_the_same_user_record_exactly_one`; `test_simultaneous_double_click_by_one_director_approves_once`
Bad amounts rejected	`test_unusable_amounts_are_rejected_with_a_message_and_not_stored`

B3. Mutation tests (about 20 seconds, needs WSL)

```
wsl -d Ubuntu-22.04 -- bash scripts/run_mutation.sh check
Remove-Item docs\mutation\check.txt
```

Expected: `Mutation score: 100.0%`, 132 mutants, 132 killed, 0 survived. Before the feature the score was 63.9% (23 of 36 killed); see document 08 for the explanation.

B4. End-to-end test with Playwright (about 15 seconds)

Make sure nothing is running on port 5000 first, then:

```
cd e2e
npm install
npx playwright install chromium
npm test
cd ..
```

Expected: `1 passed`. To watch the browser: `npx playwright test --headed`. The test signs in as the Manager, presses Approve on the 75,000.00 invoice, expects the refusal message, and checks the invoice is still pending. It starts its own server and stops it afterwards.

5. Way C: try the cases by hand in the UI

Start the app fresh (`Ctrl+C`, then start it again) so the numbering below matches.

C0. Create the test invoices once

1. Sign in as `demo.director` (password `ChangeMe123!`).
2. On the **Invoices** tab, fill **New invoice** (Customer name, Amount) and press **Create invoice**. The row appears as **Draft**.
3. Press **Submit for approval** in that row. It becomes **Pending Approval**.
4. Repeat for each invoice in this table. On a fresh server the ids are assigned in this order.

Create this	Customer name (any)	Amount	Expected id
I1	Case I1	50000.00	INV-1006
I2	Case I2	50000.01	INV-1007
I3	Case I3	500000.00	INV-1008
I4	Case I4	500000.01	INV-1009
I5	Case I5	500000.01	INV-1010
I6	Case I6	500000.01	INV-1011
I7	Case I7	500000.01	INV-1012
I8	Case I8	100.00	INV-1013

5. Press **Sign out**.

To act as a user: sign in, open the **Approvals** tab, and press **Approve** or **Reject** on the invoice. A red message appears after a refused attempt.

C1. The cases

#	Steps	Expected on screen
M1	`demo.manager` : Approve **INV-1006** (50,000.00)	Leaves the pending list; Invoices tab shows **Approved**
M2	`demo.manager` : Approve **INV-1007** (50,000.01)	Red message "A Manager may approve invoices up to 50,000.00; this invoice is 50,000.01 and needs a Director."; still pending
M3	`demo.director` : Approve **INV-1007**	Approved
M4	`demo.manager` : Approve **INV-1008** (500,000.00); then `demo.director` : Approve it	Manager refused (same Manager-limit message); Director approves alone, Approved with no "1 of 2"
M5	`demo.admin` : Approve **INV-1013** (100.00)	Red message "Role Admin is not allowed to approve invoices."; still pending
M6	`demo.director` : Approve **INV-1009**	Stays in the list and shows **1 of 2 approvals**
M7	Same Director: Approve **INV-1009** again	Red message "demo.director has already approved this invoice; invoices above 500,000.00 need two different approvers."; still 1 of 2
M8	`demo.manager` : Approve **INV-1009**	Red message "The second approver on an invoice above 500,000.00 must be a Director; demo.manager is a Manager."; still 1 of 2
M9	`demo.director2` : Approve **INV-1009**	Approved (this also shows a refused attempt does not block a valid one)
M10	`demo.manager` : Approve **INV-1010**; then `demo.director` : Approve it	First shows 1 of 2; then Approved
M11	`demo.manager` : Approve **INV-1011**; then `demo.manager` : Approve it again	First shows 1 of 2; second is refused with the duplicate-user message
M12	`demo.director` : Approve **INV-1012** (1 of 2); `demo.admin` : type a reason, press **Reject**	Rejected; its first approval stays on record
M13	`demo.manager` : on **INV-1013**, press **Reject** with the reason box empty	A message says a reason is required; still pending
M14	Same invoice: type a reason (for example "Duplicate"), press **Reject**	Rejected

#	Steps	Expected on screen
M15	Open **INV-2001** (75,000.00) as `demo.manager` and approve	Refused (this is the case the Playwright test automates)

C2. Cases you cannot do by hand (and where they are tested)

Case	Why not in the UI	Tested in
Same user with different capitals or spaces (`Dir.One` versus `dir.one`)	Usernames come from the fixed demo accounts	Policy and service unit tests (new-side and stored-side variants)
Two different Managers on a 500,000.01 invoice	There is only one Manager demo account	`test_two_different_managers_cannot_approve_above_the_director_limit`
Approve a Draft, an unknown id, or an already Approved invoice	The UI lists only pending invoices	Script cases C15 to C17; service and API tests
Delete an invoice; unusable amounts (0, negative, sub-cent, `1e999999` , NaN)	There is no Delete button, and the amount box rejects bad input in the browser	Script cases C23 and C24; API tests
Two simultaneous approvals	Needs real concurrency	Race tests (threaded) and the 4-click API test

6. Checking an API call yourself (optional)

PowerShell, with the server running on port 5000:

```
$login = Invoke-RestMethod -Method Post -Uri http://localhost:5000/api/auth/login -ContentType 'application/json' -Body '{"username":"demo.manager","password":"ChangeMe123!"}'
$headers = @{ Authorization = "Bearer $($login.token)" }
try { Invoke-RestMethod -Method Post -Uri http://localhost:5000/api/approvals/INV-2001/approve -Headers $headers }
catch { $_.ErrorDetails.Message }
```

The last line prints the refusal: `{"error":"A Manager may approve invoices up to 50,000.00; this invoice is 75,000.00 and needs a Director."}`.

7. Troubleshooting

Problem	Fix
`pip install` or imports fail	Run the install command from the repository folder; use Python 3.10 or newer
"Address already in use"	Another server is on that port. Stop it or use `--port 5001`
UI shows old data or INV numbers differ from this guide	Restart the server for a fresh start
A manual try finds INV-2001 already approved	An earlier run approved it. Restart the server to reset the data. (The Playwright test creates its own invoice and passes regardless.)
Mutation run fails from Git Bash	Use PowerShell, or prefix `MSYS_NO_PATHCONV=1`
`try_all_cases.py` cannot start its server	Check nothing uses port 5058, or use `--port 5059`, or point it at a running server with `--url`
A case fails	Read the line under it starting with `->`: it shows the status and body received. Then open the matching test in the table in section 4

8. Quick checklist

- [ ] `python scripts/try_all_cases.py` gives `25 of 25 cases passed`
- [ ] `python -m pytest backend/tests -q` gives `177 passed`
- [ ] Mutation run gives `100.0%`
- [ ] `npm test` in `e2e` gives `1 passed`
- [ ] You tried M1 to M15 by hand (or at least M2, M7, M8, M9)